

L36: Model Credibility and Documentation

Simulation Without a Black Box

Outline

What Is Credibility?

Sargent's V&V Diagram

Documentation Requirements

Simulation Report Structure

Communicating Uncertainty

Reproducibility and Version Control

Professional Standards

Key Takeaways

Credibility: Trust from Intended Users

Definition

A model is **credible** when the intended decision makers trust its outputs sufficiently to act on them — not when it is “absolutely correct.”

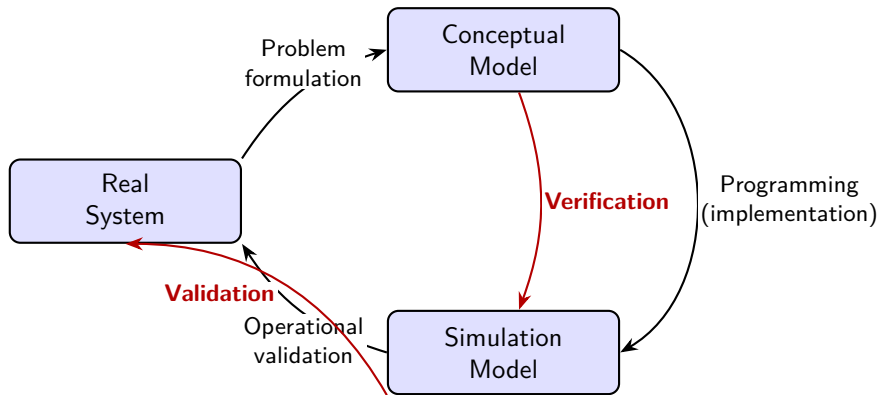
Credibility is built from:

- Transparent assumptions
- Documented V&V evidence
- Honest statement of scope and limitations
- Reproducibility (same seed $\Rightarrow$ same output)
- Peer review or expert sign-off

Danger

Credibility can be destroyed in one meeting if an expert asks a simple question (e.g., “What happens if demand doubles?”) and you have no answer. Sensitivity analysis must be done *before* presenting results.

Sargent's V&V Framework



Adapted from

R. G. Sargent, "Verification and Validation of Simulation Models," 2013.

Documentation Requirements

Conceptual model document

- System boundary diagram
- Entity types, resources, queues
- Routing logic and decision points
- All assumptions and simplifications

Data sources

- Where each distribution came from
- Sample size and collection period
- Goodness-of-fit test results

V&V evidence package

- Verification checklist (signed off)
- Unit test results (pytest output)
- Extreme-condition results
- Historical comparison table
- Face validity sign-off by expert

Warning

Documentation written *after* the fact is less reliable than documentation written concurrently. Use version-controlled notebooks.

Simulation Report Structure

Section	Content
Executive Summary	Decision question, key finding, recommended action
Model Description	Conceptual model, assumptions, scope
Input Data	Distributions fitted, sources, GoF tests
V&V Evidence	Verification checklist, benchmark comparisons, validation results
Results	KPI tables, CI widths, scenario comparisons
Sensitivity Analysis	What happens if key inputs change by $\pm 20\%$?
Limitations	What the model <i>cannot</i> answer; validity scope
Appendices	Code listing, full parameter table, raw data

Communicating Uncertainty to Decision Makers

What a 95% CI Means

“If we were to run 100 independent sets of replications under identical conditions, approximately 95 of the resulting intervals would contain the true mean.”

It is **not** “the true value is in this range with 95% probability.”

Practitioner framing: “Our estimate of mean wait is 12.4 minutes. The simulation uncertainty (± 0.8 min) does not change the recommendation — alternative B is consistently better across all replications.”

Common Mistake

Reporting a single-replication point estimate with no interval.
Practitioners interpret it as exact — it is not.

Practical Rule

Always show:

- Point estimate
- Half-width of 95% CI
- Number of replications n

Reproducibility: Seed, Versions, and Checksums

Every published result must be reproducible:

1. Fix random seeds — pass as constructor arguments.
2. Pin package versions (requirements.txt or pyproject.toml).
3. Checksum input data files (sha256sum data/clinic.csv).
4. Archive the exact code version (git tag or commit hash).

```
# report_metadata.py
import simdes, simpy, numpy, scipy
print({
    "simdes":  simdes.__version__,
    "simpy":   simpy.__version__,
    "numpy":   numpy.__version__,
    "scipy":   scipy.__version__,
    "seed":    42,
    "data_sha": "a3f7b2...",
})
```

Embed this block in every report notebook.

Professional Standards: INFORMS Simulation Best Practices

Selected INFORMS / WSC guidelines:

- Model purpose and scope must be stated explicitly.
- All assumptions must be documented and reviewed.
- V&V must be conducted and reported.
- Output analysis must use proper statistical methods (not single run).
- Sensitivity analysis is mandatory for high-stakes decisions.
- Results must be presented with appropriate uncertainty.

Model Limitations Statement

Every report must include:

“This model is valid for [conditions].
It does not represent [exclusions].
Results should not be extrapolated to
[out-of-scope scenarios].”

Stating limitations is not a weakness
— it is professionalism.

Key Takeaways

L36 Summary

1. Credibility is trust from intended users, not absolute correctness — it must be earned through transparent documentation.
2. Sargent's diagram maps three relationships: problem formulation, verification, and operational validation.
3. A complete simulation report includes: model description, data sources, V&V evidence, results with CIs, and an explicit limitations section.
4. Communicate CIs as precision statements, not probability ranges.
5. Reproducibility requires fixed seeds, pinned package versions, and checksummed input data — log all three in every report.

Next: L37 — Simulation-based optimization.